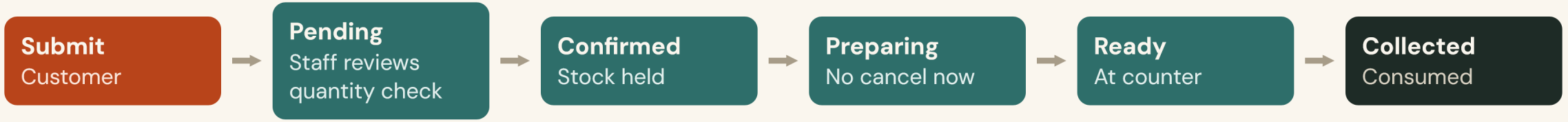
01 · PROBLEM UNDERSTANDING

stock, per slot, under contention

One lifecycle where a slot can never be oversold, and every refusal tells the customer why.

Workflow we prioritised: one order, end to end



Rejected (staff, with reason) or **Cancelled** (customer, before prep) → stock released at once. Collected stays consumed.

Who it serves

Customer: available items, order, track, cancel

Outlet Staff: confirm or reject, move orders to pickup

Outlet Manager: slot quantities, flag shortages + reason

Secondary: campus admin

17 rules from 12 conditions

Available = capacity - held by active orders

All-or-nothing: one short line fails the order

Only Rejected or Cancelled release stock

No cancel once Preparing; no second collect

Flagged unavailable blocks new orders only

Assumptions and 90-min scope

One outlet, no login: role selector (A1, A4)

Cancel allowed while Pending too (A6)

App clock 4 Aug 2026 10:00 for cut-offs (A3)

In: full lifecycle × 3 role views, seed + reset

Out: payments, notifications, edits, partial fills

02 · SOLUTION AND ENGINEERING DECISIONS

Rules in one layer, UI stays thin

ui → services → core. Dependencies point one way only.

Bonus · In-app notifications
Each status change lands in the customer's inbox, in the same transaction

`ui/`
4 role views; only buttons from allowed_actions() appear

→

`services/`
ordering, lifecycle, availability, menu, clock, notify

→

`core/`
TRANSITIONS table, tx() with BEGIN IMMEDIATE, schema, seed

Design decisions

Availability derived from COUNTED orders, never stored

One TRANSITIONS table drives every button

BEGIN IMMEDIATE around every write

Hybrid confirm: small orders auto, the rest to staff

Trade-offs

Reserve at submit: pending orders may briefly hold stock

Pickup code, not login, for cancel and collect

Plain functions + sqlite3: no ORM, no classes

Fresh src/ build over patching the old backend

Validation and team

AT-01 to AT-18 as pytest, plus edge cases

Concurrency: two buyers, last unit, one wins

AI_LOG.md records what we caught in AI output

Split by file: A ordering, B lifecycle, C UI

Limits → next

No real login → **campus SSO and roles**

Stuck orders hold stock → **auto-release timeout**

Refresh, not live → **push alerts**

Admin demand view → **event forecasting**